SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

OFFICIAL COURSE SYLLABUS & LEARNING CONTRACT

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Course Status: Core / Elective Undergraduate | Credit Rating: 12 Credits | Total Notional Hours: 120 Hours
Hours Distribution: 15h Lectures, 30h Seminars/Tutorials, 15h Assignments, 15h Independent Study, 45h Practical Laboratory

1. Course Aim & Strategic Rationale

BIT 111: Microcomputers and Applications is designed to empower undergraduate students across various engineering, agricultural, applied sciences, and environmental disciplines with foundational knowledge and applied practical competence in modern microcomputer systems. In an era increasingly driven by Agriculture 4.0, data-driven farm enterprise management, smart sensors, and automated information processing, microcomputers serve as the ubiquitous tool for computational analysis, scientific documentation, electronic telemetry, and institutional workflow automation. This course establishes both the physical hardware foundations of personal computing and the hands-on proficiency required to operate operating systems, conduct advanced word processing, build dynamic quantitative spreadsheets, browse securely, and defend digital assets against cyber threats.

2. Course Expected Learning Outcomes (CELOs)

Upon successful completion of BIT 111, each student will be able to:

1. Identify and describe the principal hardware components of a modern microcomputer system and explain their physical and logical interconnects.
2. Utilize modern and legacy operating systems (including command-line DOS and graphical Windows/Linux environments) to manage processes, files, directories, and peripheral devices.
3. Configure and apply personal computer (PC) security controls, including power-on passwords, BIOS/UEFI setup parameters, Secure Boot, TPM encryption, and multi-factor operating system login policies.
4. Demonstrate advanced proficiency in word processing application software (Microsoft Word / LibreOffice Writer) to produce professional academic reports, automate multi-level styling, implement mail merges, and manage citations.
5. Construct dynamic computational spreadsheet models (Microsoft Excel / LibreOffice Calc) applying mathematical, statistical, lookup, and financial functions, conditional formatting, pivot tables, and analytical visualizations for agricultural and scientific decision-making.
6. Apply computer application software and security countermeasures (including 3-2-1 backup strategies, anti-malware scanners, and email protocol netiquette) to resolve authentic real-world data management challenges.

3. Course Structure & Syllabus Modules

Topic #	Curriculum Topic Title	Detailed Syllabus Subtopics & Practical Competencies	Notional Hours
Topic 1	Introduction to Computers	1.1 Computer Generations (1st to 5th+ Gen: vacuum tubes, transistors, ICs, microprocessors, AI/quantum); 1.2 History & Pioneers (Abacus, Babbage, Lovelace, ENIAC, von Neumann, microcomputer revolution); 1.3 Computer Classifications (Size: micro, mini, mainframe, supercomputer; Technology: analog, digital, hybrid; Purpose & network roles).	15 Hours
Topic 2	Main Components of a Microcomputer	2.1 Devices & Logical Configurations (CPU: ALU, CU, Registers, Cache; Bus architecture: Data, Address, Control; System RAM DDR4/DDR5, ROM/UEFI, NVMe SSD, HDD; Input/Output devices, Motherboard chipsets); 2.2 Comparative Architecture (Micro vs Mainframe, Workstations vs Micro, Minicomputers vs Microcomputers).	20 Hours
Topic 3	PC Security Controls	3.1 Hardware & Power-on Security (Hard disk ATA passwords, Power-On Password, Chassis intrusion); 3.2 Firmware & System Setup Security (BIOS/UEFI Supervisor & User passwords, Secure Boot, TPM 2.0, BitLocker / LUKS drive encryption); 3.3 OS Login & Access Controls (Windows Hello, PINs, Passwords, User Account Control, Least Privilege).	15 Hours
Topic 4	Functions & Types of Operating Systems	4.1 Evolution & OS Categories (Single/Multi-user, Multitasking, RTOS, Batch; MS-DOS, UNIX/Linux, Windows); 4.2 Disk Operating System (DOS Architecture, COMMAND.COM, Internal vs External commands, batch automation); 4.3 Graphical Operating Systems (Kernel architecture, Memory management, Paging, NTFS/ext4, Admin tools).	20 Hours
Topic 5	Microcomputer Applications: Word, Excel, Web & Email	5.1 Word Processing (Templates, Styles, Multi-level numbering, Automated Table of Contents, Tables, Mail Merge); 5.2 Spreadsheet Modeling (Relative, Absolute, Mixed cell references, Formulas: SUM, AVERAGE, IF, VLOOKUP/XLOOKUP, Charts, Pivot Tables); 5.3 Web Browsing & Email (HTTP/HTTPS, DNS, Web browsers, SMTP, IMAP, POP3, Netiquette, Spam filtering).	30 Hours
Topic 6	Data Security & Malware Defense	6.1 Physical Protection & Environmental Controls (UPS, Surge protectors, 3-2-1 Backup rule, Disaster recovery); 6.2 Logical Access Control & Cryptography (Authentication vs Authorization, ACLs, Symmetric AES vs Asymmetric RSA); 6.3 Malware Mitigation (Viruses, Worms, Trojans, Ransomware, Spyware, Rootkits, Anti-malware scanners).	20 Hours

4. Teaching and Learning Methodology

i. Interactive Lectures & Multimedia Presentations: Interactive face-to-face lectures and conceptual demonstrations covering computer hardware architecture, operating system design principles, and cybersecurity foundations.
ii. Hands-on Laboratory Practical Sessions: Intensive bench sessions in the university computer laboratory: exploring BIOS/UEFI setup screens, executing command-line DOS batch utilities, configuring drive security, building multi-page formatted technical reports in Word, creating complex agricultural data models in Excel, and configuring anti-malware software.
iii. Practical Demonstrations & Problem-Solving: Live demonstration of formula auditing, spreadsheet error troubleshooting, mail merge configuration, and network email client setup.
iv. Self-Directed Takehome Homework Assignments: Six comprehensive tiered assignments reinforcing conceptual understanding, quantitative calculations, command syntax, and real-world enterprise scenarios.

5. Assessment Methods & Grading Policy

Assessment for BIT 111: Microcomputers and Applications consists of Continuous Assessment (60%) and a comprehensive University Examination (40%). Practical hands-on competence in Word, Excel, DOS, and system security is heavily weighted to ensure true technical operational proficiency.
Note: Specific weighting allocations may be adjusted by the course instructor in accordance with departmental guidelines.

Assessment Component	Scope / Description	Weighting (%)	Delivery Mode / Deliverable
Two Takehome Assignments	Selected from the 6 Course Topic Homeworks	5%	Individual analytical and practical submissions (2.5% each)
Two Theory Tests	Theory Test 1 (Topics 1–3) & Theory Test 2 (Topics 4–6)	15%	Closed-book written engineering theory examinations (7.5% each)
Practical Laboratory Report	Formal logbook of laboratory practical sessions (Word, Excel, DOS, Security)	10%	Submitted laboratory logbook with verified output screenshots and analysis
Practical Test	Hands-on examination in the Computer Lab (Word Processing, Excel Modeling & DOS)	30%	Timed hands-on laboratory examination testing live application proficiency
Continuous Assessment (CA) Subtotal	Assignments (5%) + Theory Tests (15%) + Lab Report (10%) + Practical Test (30%)	60%	Total In-Semester Continuous Assessment Score
University Examination (UE)	Comprehensive Final Examination covering Topics 1 through 6	40%	Formal 3-hour theoretical and applied university examination
Total Final Grade	Continuous Assessment (60%) + University Examination (40%)	100%	SUA Grading Scale: A (70%+), B+ (60-69%), B (50-59%), C (40-49%), D (35-39%), E (0-34%)

6. Prescribed Textbooks & Reference Literature

1. Shelly, G. A., & Vermaat, M. E. (2018). Discovering Computers: Digital Technology, Data, and Devices (16th ed.). Cengage Learning.

2. Parhami, B. (2005). Computer Architecture: From Microprocessors to Supercomputers. Oxford University Press.

3. Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). Operating System Concepts (10th ed.). John Wiley & Sons.

4. Walkenbach, J. (2019). Microsoft Excel 2019 Bible: The Comprehensive Tutorial Resource. Wiley.

5. Stallings, W. (2021). Computer Organization and Architecture: Designing for Performance (11th ed.). Pearson.

6. Young, M. L., Barrows, A., & Stockman, J. C. (2010). Access 2010 ALL-IN-ONE For Dummies. Wiley Publishing Inc.

7. Tanenbaum, A. S., & Bos, H. (2015). Modern Operating Systems (4th ed.). Pearson Education.

8. Forouzan, B. A. (2013). Data Communications and Networking (5th ed.). McGraw-Hill Education.

9. Castellan, N. J. (1988). Comments on applications of microcomputers in teaching. Behavior Research Methods, Instruments, & Computers, 20(2), 193-196.